SELECTED USE CASE / REAL-TIME INTELLIGENCE

Battery Returns Intelligence

HACKATHON USE CASE - REAL-TIME ACTIONS

Turning Customer Real-Time Data into Real-Time Agentic Actions

PROBLEM STATEMENT

Disconnected return records hide who initiated a return, what inspection found, and when a batch needs attention.

OUR CHOSEN SCENARIO

Battery-return spike

> detect batch pattern

> investigation notification

Delivered: live detection + Teams alert | Agentic loop: partial, not autonomous

Where each use-case section is implemented.

Read the brief as data > intelligence > action; distinguish delivered functions from supporting features and gaps.

USE-CASE SECTION	PROJECT PART	WHAT IS IMPLEMENTED	STATUS
Real-time customer data	Eventstream / Eventhouse	Synthetic live return events ingested	**DEMO**
Real-time intelligence	KQL BatchSignals	3 distinct batteries / batch / 15 minutes	**VERIFIED**
Real-time action	Activator + Teams	An investigation notification was delivered	**VERIFIED**
Agentic actions	AI companion	Read-only, user-initiated; no automatic ticket	**PARTIAL**
Trusted context (support)	Lakehouse pipeline	Bronze / Silver / Gold; 12,000 historical cases	**BUILT**
Experience / cost (extra)	Dashboards	Fabric visuals, separate local AI, cost estimates	**BUILT**

BRIEF'S EXAMPLES - OUR PROJECT ADAPTS THE INVESTIGATION PATTERN

Machine anomaly > maintenance | Inventory drop > replenishment | Shipment delay > operations

Suspicious transaction > investigation | Network issue > service action

Two data paths. One investigation experience.

A repeatable file pipeline builds trusted history; an independent live path supports investigation alerts.

TRUSTED HISTORICAL CONTEXT - SUPPORTS THE USE CASE

4 source files

3 CSV + 1 JSON

Products, batteries, shipments, return events

>

BRONZE

Preserve raw values and source provenance

>

SILVER

Clean and validate

Quarantine bad events

Remove exact replays

>

GOLD

Join dimensions

Latest revision per return

Publish after validation

>

EXPLORE

Eventhouse + dashboard

Separate AI companion

REAL-TIME DATA > INTELLIGENCE > AUTOMATED ACTION

Eventstream

Receives live battery-return events

>

Eventhouse

Queries the live event population

>

Activator + Teams

Sends an investigation notification

Trusted data, with checks you can explain.

SUPPORTING CONTEXT: a fixed historical snapshot, not the live alert population.

46,442

SOURCE EVENTS

Raw landing events preserved

821

EXACT REPLAYS REMOVED

Duplicate events, not repeat returns

3

INVALID EVENTS QUARANTINED

Rejected by validation rules

12,000

GOLD RETURN CASES

One highest-revision row per return

What we verified

45,618 valid, distinct events

11,693 distinct batteries

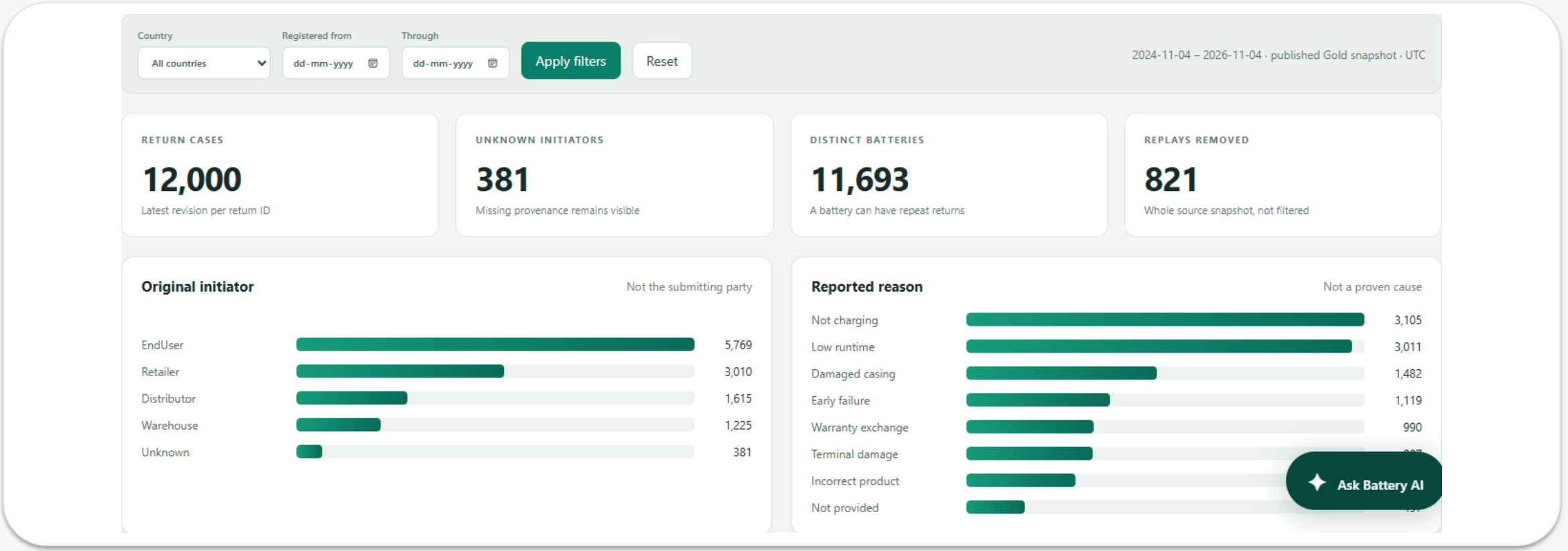
Legitimate repeat returns retained

Rerun: no duplicate publication

Reliable numbers first. Visuals and AI second.

Explore the return story in one place.

SUPPORTING VIEWS: local-companion screenshot; the Gold dashboard is historical, not live.



12,000 cases | 381 unknown initiators | Country and date filters

Ask naturally. Verify the answer.

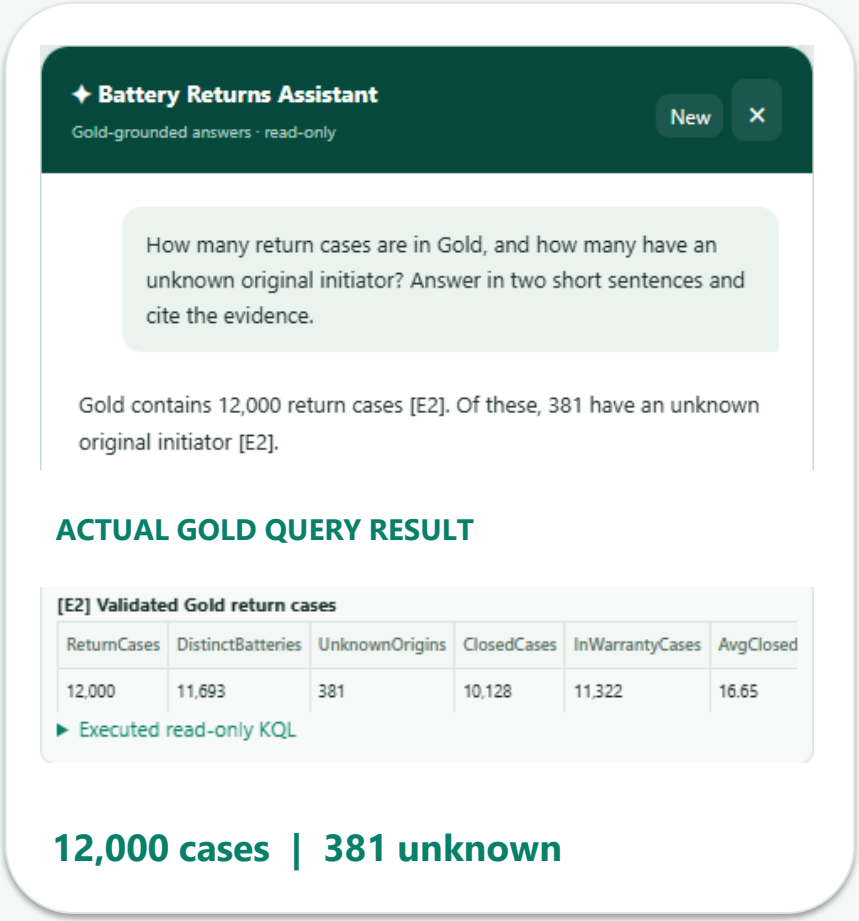
AGENTIC ACTIONS - PARTIAL: read-only, user-initiated AI; no autonomous ticket or remediation.

ASK "How many cases have an unknown original initiator?"

GROUND The assistant retrieves actual data, rather than inventing a number.

VERIFY Open the evidence to inspect the executed query and returned rows.

It does not invent defect rates when the units-sold denominator is missing.



Move from insight to investigation.

REAL-TIME ACTION: delivered Teams notification. Capacity estimates are an extra feature.

investigation alert

Live events > Eventhouse > Activator > Teams

Demo rule: 3 distinct batteries

from the same batch within 15 minutes.

Teams notification received in the demo.

This threshold is not a validated business policy.

Extra: capacity estimate

Example: F2 / USD / 24 charged hours per day

1 DAY	1 WEEK
$9.60	$67.20
30 DAYS	**365 DAYS**
$288.00	$3,504.00

Current workspace: FT1 trial. These are estimates, not actual bills.

West Central US retail snapshot, Nov 4, 2026. Capacity only; excludes storage, licenses, taxes, bandwidth and other services.

Demonstrate the use-case coverage honestly.

Show the live alert path separately from historical Gold and read-only AI.

1	**Show live-path evidence**	event ingestion, batch rule and received Teams notification.
2	**Trace historical cases**	Gold + Fabric dashboard: filters and case evidence.
3	**Ask the AI companion**	read-only analysis; verify the query and returned rows.
4	**State the remaining gap**	No automatic AI investigation, ticket creation or remediation yet.

Open original Fabric dashboard

Open local AI companion (this PC)